



PERTH MODERN SCHOOL

Exceptional schooling. Exceptional students.

INDEPENDENT PUBLIC SCHOOL

Semester One Examination, 2019

Question/Answer booklet

MATHEMATICS

UNIT Methods 1 & 2

Section One:
Calculator-free

Name: _____

Teacher's Name: _____

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Question	Mark	Question	Mark
1		6	
2		7	
3		8	
4		9	
5		Total	/52

MATHEMATICS METHODS 1 & 2**CALCULATOR-FREE****Structure of this paper**

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	9	9	50	52	36
Section Two: Calculator-assumed	14	14	100	92	64
Total					100

Instructions to candidates

1. The rules for the conduct of the Western Australian Certificate of Education ATAR course examinations are detailed in the *Year 12 Information Handbook 2016*. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your answers to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Additional pages for the use of planning your answer to a question or continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
5. **Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you **do not use pencil**, except in diagrams.
7. The Formula sheet is **not** to be handed in with your Question/Answer booklet.

Section One: Calculator-free**(52 Marks)**

This section has **nine (9)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Working time: 50 minutes.

Question 1**(5 marks)**

Solve the following

(a) $\frac{5x}{6} = \frac{7}{2}$ (1 mark)

(b) $\frac{3x}{5} - \frac{x}{4} = 2$ (2 marks)

(c) $x^3 + 3x^2 - 4x = 0$ (2 marks)

MATHEMATICS METHODS 1 & 2**CALCULATOR-FREE****Question 2****(12 marks)**

Brodie and his sister Ryanne are out playing fetch with their Dog, on a straight slope. The three are spread out evenly with the Dog being directly (half-way) in between the Brodie and Ryanne.

When modelled on a cartesian plane, the Dog's position can be considered as (7.5, 2.5) and Ryanne's position can be considered as (10, 3).

(a) Determine the coordinates of Brodie. (2 marks)

(b) Determine the equation of the slope. (3 marks)

(c) Brodie is flying a kite. If the line of the kite is running perpendicular to the slope, determine the equation that models the line. (3 marks)

(d) Ryanne throws the ball away from the Dog and the path of the ball can be modelled by the following quadratic equation.

$$y = -\frac{x^2}{2} + x + 1$$

Determine the coordinates that the ball first lands on the slope. (4 marks)

Question 3**(3 marks)**

Find in terms of p the remainder when $x^3 - 2x^2 + px - 6$ is divided by $x - 2$.

Question 4**(3 marks)**

Points $M(p, 6)$ and $N(1, -2)$ are the end points of line segment MN . If the midpoint of line is $(-3, q)$. Determine the value of p and q .

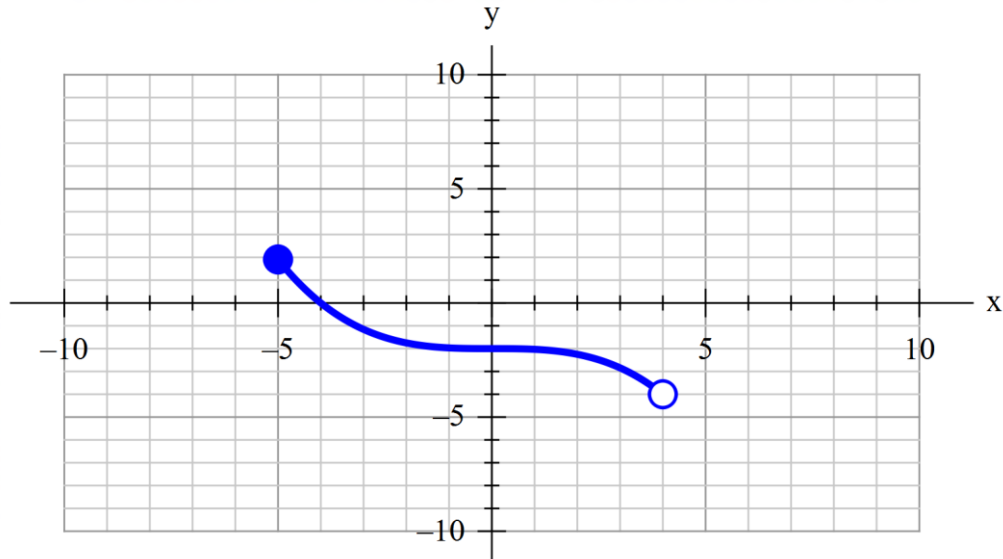
MATHEMATICS METHODS 1 & 2

CALCULATOR-FREE

Question 5

(6 marks)

The function $y = f(x)$ is shown below.



(a) State the range of $f(x)$.

(1 mark)

(b) Another function is given by $g(x) = 2f(x - 3)$.

(i) Describe the transformation required to produce $g(x)$ from $f(x)$.

(2 marks)

(ii) State the coordinate of the y-intercept under this transformation.

(1 mark)

(c) On the same axes above, sketch the graph of $y = f(2x) + 2$.

(2 marks)

Question 6**(7 marks)**

Given the equation $2^{x+1} + 2^{3-x} = 17$, $x \in R$,

- (a) use a suitable substitution to rewrite the equation above as a quadratic equation, (3 marks)

- (b) hence, determine the solution(s) to the exponential equation. (4 marks)

MATHEMATICS METHODS 1 & 2

CALCULATOR-FREE

Question 7

(6 marks)

Given $f(x) = ax(x^2 + 4x + 3)$, $-3 < f(x) \leq 0$ and $f(1) = -24$

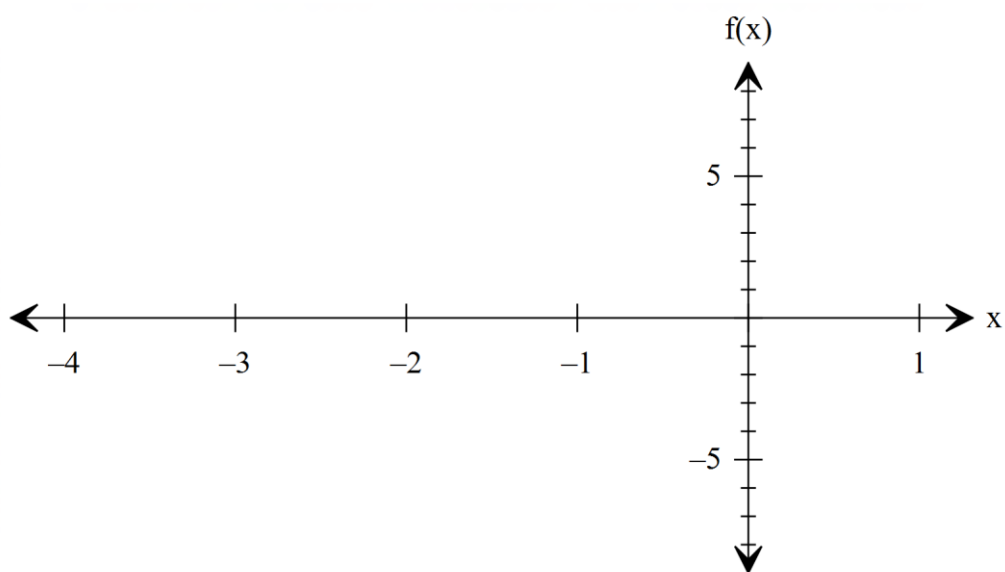
(a) Rewrite the function in factorised form, stating the value of a .

(3 marks)

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(b) Sketch the function on the given domain.

(3 marks)



Question 8

(6 marks)

A relationship between x and y is given by $x = \frac{7}{2}y + 2$.

(a) Determine y when $x = 23$. (2 marks)

(b) State, with justification, whether x is a function of y . (2 marks)

(c) State, with justification, whether x is proportional to y . (2 marks)

Question 9

(4 marks)

The quantity of light, Q , given out by a lamp is proportional to the square of the current, I , passing through the lamp. When $Q = 1000$, $I = 2$.

(a) Write an equation that relates Q and I . (2 marks)

(b) Find the value of Q when $I = 3$. (1 mark)

(c) What happens to Q when I is doubled. (1 mark)

MATHEMATICS METHODS 1 & 2**CALCULATOR-FREE****Additional working space**

Question number:

CALCULATOR - FREE

MATHEMATICAL METHODS 1 & 2

Additional working space

Question number:

